

IMO(7th) – 1965

First Day

- Find all real numbers $x \in [0, 2\pi]$ such that $2 \cos x \leq \left| \sqrt{1 + \sin 2x} - \sqrt{1 - \sin 2x} \right| \leq \sqrt{2}$.
- Consider the system of the equations:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = 0 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = 0 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = 0 \end{cases}$$

- a_{11}, a_{22}, a_{33} are positive real numbers;
- all other coefficients are negative;
- in each of the equations the sum of the coefficients is positive.

Prove that $x_1 = x_2 = x_3 = 0$ is the only solution to the system.

- A tetrahedron ABCD is given. The lengths of the edges AB and CD are a and b respectively, the distance between the lines AB and CD is d, and the angle between them is equal to w . The tetrahedron divided into two parts by the plane π parallel to the lines AB and CD. Calculate the ratios of the volumes of the parts if the ratio between the distances of the plane π from AB and CD is equal to k.

Second Day

- Find all sets of four real numbers x_1, x_2, x_3, x_4 such that the sum of any of the numbers and the product of the other three is equal to 2.
- Given a triangle OAB such that $\angle AOB = \alpha < 90^\circ$, let M be any arbitrary point of the triangle different from O. Denote by P and Q the feet of the perpendiculars from M to OA and OB respectively. Let H be the orthocenter of the triangle OPQ. Find locus of the point H when:
 - M belongs to the segment AB.
 - M belongs to the interior of triangle OAB.
- We are given $n \geq 3$ points in the plane. Let d be the maximal distance between two of the given points. Prove that the number of pairs of points whose distance is equal to d is less than or equal to n.

**